

WHAT MACHINE COGNITION MAKES POSSIBLE

Ad hoc semantic models.

When two systems can **reason**, each can lift its own data into a complete semantic model — which they can then reconcile, machine-to-machine, with nothing agreed in advance.

THE QUESTION

Do two cognitive systems still need a data model agreed in advance to work together?

Or can they build a good-enough one **on their own**, for the job at hand?

*Decades of legacy systems that can't explain themselves, data of mixed quality, operations that must never break, all real. The question is whether the pre-agreement, the slow and expensive part, will still be required in a world of increasingly **cognitive** systems, and in what form.*

The answer we reach: yes, with real qualifications

THE CENTRAL OBJECT

One object sits at the centre: a complete semantic model.

Everything turns on the model a system's data is **lifted** into. It is **semantic**: it carries meaning, not just structure; **ad hoc**: built for the case at hand, with no standard agreed in advance; and **complete**.

Complete the **ontology**, its concepts and their instances (TBox and ABox), **plus** the two layers a bare schema never carries: **pragmatics** (what a thing is *for*, whose judgement governs it) and **provenance** (who asserted it, and how firmly).

*It is the **output of the lift**, and the **input to reconciliation**.*

THE LIFT

Producing the semantic model is a cognitive act, with a bar to clear.

The **lift**, turning a system's data into that complete model, is the first thing we ask machine cognition to do.

The bar it must clear: a lift is **adequate** when a *sufficiently capable* cognitive element can pick the model up and apply it to **any** end, from the package alone. A model that clears the bar is **portable**. There are real limits, and assists, to the lift processes to achieve portability.

The assist: a reference a “standard” vocabulary (a lexicon, not an ontology or data schema) a lift or reconciliation can tie terms and meanings to.

A well-lifted model is understood cold: by strong, middle, and even weak readers.

Hand a lifted model to an independent cognitive agent that never saw the system; ask it to explain each concept from the package alone. We measure comprehension directly, not in executing any particular operational task with the model.



comprehension on the hardest no-standard scenario

Holds across six independent cases, and across model families: a strong reader from a different family reads a lift as well as one from the family used in the lift. Honest floor: weak models can confabulate.

A reference may assist model comprehension or otherwise facilitate operations that use the model, so make it, or a pointer to it, part of the model package.

Two ways a reference earns its place. It can **assist comprehension**, helping a reader make sense of the model on its own. And it can **facilitate the operations that use the model**, above all **reconciliation**, where a shared reference is common ground two models can bind through.

A careful lift can make a model comprehensible with **no reference at all**, so the first use may fall away. The operations that use the model still benefit. So, **make the reference, or a pointer to it, part of the package regardless**: it is thin, and it pays for itself downstream.

Semantic model comprehension is generally achieved whether or not the model ships with or points to a reference; but a reference may rescue a cognitively weak consumer and makes a reconciliation operation cheaper.

THE TURN

A major prize: two systems that never agreed on anything, working together.

A semantic model is an object. **Reconciliation is a flagship use of it**, aligning two independently built models so the systems that use them natively can interoperate.

Reconciliation is an operation that begins with semantic models. It is not about patching bad lifts: two sound models still diverge on **lexicon**, **ontological structure and span**, and **pragmatics**, and bridging that divergence is the reconciliation task.

RECONCILIATION HOW A REFERENCE HELPS

When no reference is available, capable agents bind what clearly matches and refuse to bind what does not.

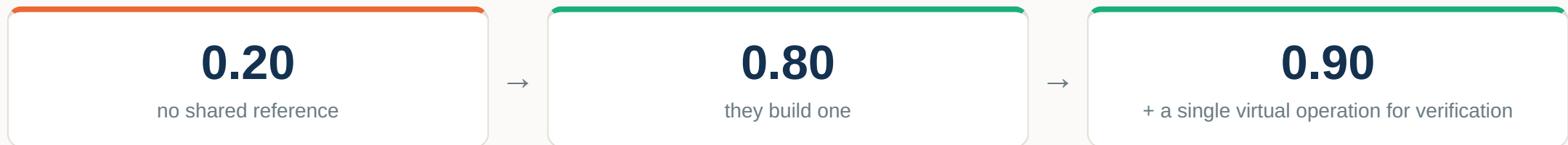
The discipline is the point: a false match (a transport *grade* vs an IP *grade*, similar names, unrelated meanings) is refused. A reference helps pick up from there.

Disciplined refusal, not naïve agreement.

When a reference isn't there, cognition can build it.

Two private models, attempting to reconcile, no shared reference. Left alone, a careful cognitive agent refuses to guess and binds almost nothing.

Let the two agents **construct a thin shared reference themselves**, then bind through it:



*The numbers are **resolved fractions**: the share of the true correspondences the agents find and commit, every one of them correct (precision stays 1.00). Below 1.0 is **deferral, not failure**: what a thin self-built reference hasn't surfaced yet. Full reconciliation's verification closes the rest to 1.0, and starting from the higher number reduces required cognitive effort. They don't need a standard handed to them; they can build one.*

RECONCILIATION THE PAYOFF

Where both sides can reason, reconciliation always completes.

Two capable cognitive systems can always achieve full reconciliation autonomously: through Q&A, virtual experiments and other verifications, etc. As cognition recedes from one or both systems, a reference assumes a larger role, and full autonomy might not be achieved by machine cognition employed in overlay to inert systems.

WHAT THE EVIDENCE SHOWS

Four scenarios, spanning different operations classes and ways semantic models diverge.

Configuration: aligning two semantic models of one network, each lifted from a standard interface schema; **intent:** negotiating a declarative wish into a concrete realisation; **standard-free:** provisioning across two private systems with no shared standard; **observability:** deciding what an observed signal means. Four operations classes, chosen so that together they also span the kinds of divergence reconciliation must bridge.

Each scenario is run on three independently built **cases:** three different pairs of **semantic models** to reconcile.

Every headline result reproduces across the independently built **cases** (different pairs of semantic models) and across three independent **AI families**. It is neither a quirk of one model pair nor of one vendor's reasoner.

WHAT THE WORK PROVES

Systems can be made to interoperate in the absence of pre-agreed data schemas: by creating ad hoc semantic models for each system, then reconciling the divergences those models may feature.

Both the semantic-model production (the lift) and the reconciliation are cognition-based acts.

Semantic model portability is real and measurably verifiable.

A capable cognitive agent can understand and use a soundly produced semantic model, directly and without other supports. We verify this across a cognitive capability ladder and independent AI families.

Capable cognition reconciles divergent semantic models on its own.

Where systems are cognitively capable, two semantic models are fully reconciled through their autonomous Q&A exchanges and virtual verification operations, even where no standard schema are in sight.

Portability and reconciliation degrade gracefully as cognition recedes, not all-or-nothing.

A reference rescues a cognitively weak consumer and makes reconciliation measurably cheaper.

What is useful to standardise may evolve, toward simpler artifacts that are more flexibly and broadly usable.

What's worth agreeing in advance may narrow and simplify: as semantic models can increasingly be produced and reconciled ad hoc, for the task at hand, the heaviest fully-specified schemas may give way to lighter, more general **references**, plus the few matters that still call for a decision: questions of authority, not of meaning. Legacy standards coexist throughout.

An evolution, not a rupture.

The full picture — what semantic models look like and how they are built, how portability has been measured, how references are best designed and how they may be used to measured effects, and how semantic models are reconciled, demonstrated and evaluated in depth across four operations scenarios — is in the report.